

Saramsh Gautam

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EDUCATION

Louisiana State University

Baton Rouge, Louisiana

M.S. and Ph. D. in Computer Science

Expected Graduation, May 2026

- o **Concentration:** Human Computer Interaction (HCI)
- o **GPA:** 4.12
- o **Related Coursework:** Algorithm Analysis and Design, Programming Language Structure, Machine Learning, Artificial Intelligence, Database Management System, Object-Oriented Programming, Big Data, Statistics & Applications, Advance Software Engineering

Tribhuvan University

Kathmandu, Nepal

Bachelor's of Engineering, Computer Engineering

Aug 2022

- o **Concentrations:** Computer Science, AI and ML
- o **GPA:** 4.00
- o **Related Coursework:** Object-Oriented Programming, Theory of Computation, Operating System, Data Structure and Algorithm

SKILLS

Programming:

Python, JavaScript, HTML/CSS, SQL, Node.js, React.js, C++, C

Tools & Platforms:

VS Code, Git, Jupyter Notebooks, Docker, AWS, Firebase, NGINX, Linux Server Deployment, Hugging Face Hub

Frameworks & Libraries:

Flask, FastAPI, Spring Boot, TensorFlow, PyTorch, OpenAI API, Transformers, LoRA, LangChain

Data & Visualization:

D3.js, Chart.js, Streamlit, Pandas, NumPy, Matplotlib, Seaborn

Machine Learning & NLP:

Whisper, LLaMA, Transformers, Retrieval-Augmented Generation (RAG), Instruction Tuning, ROUGE Evaluation

Research & HCI Methods:

Thematic Analysis, User Study Design, Qualitative Coding, Survey and Study Design, Human Studies Protocol

EXPERIENCE

Louisiana State University

Baton Rouge, Louisiana

Research Assistant

Aug 2023 – Present

- **PolyFlux:** Develop an AI-facilitated infinite-canvas brainstorming tool (TLDraw/React + Flask) with adaptive nudge to support divergent and convergent ideation.
- **Multilingual Collaboration Systems:** Co-developed LINC, a tool for real-time multilingual transcription and post-meeting analysis to support ESL students in academic collaboration.
- **Real-Time Transcription Pipeline:** Integrated Whisper and WebSockets with Python Flask for low-latency audio transcription in live meetings.
- **Data Visualization for Reflection:** Built stream graphs, radar charts, and chord diagrams in React/D3.js to visualize speaking patterns and comprehension dynamics.
- **Full-Stack Development:** Developed end-to-end infrastructure using Flask microservices and React frontend with Firebase for persistent data handling.

- **DevOps and Deployment:** Deployed the system on university Linux servers with NGINX and SSL for secure access.
- **Qualitative Research & Thematic Analysis:** Conducted formative studies with ESL researchers, leading survey/interview design, participant recruitment, and data collection. Performed thematic analysis through iterative coding to uncover communication challenges and tool usage strategies.
- **Research Focus and Growth:** Developed expertise at the intersection of qualitative methods and AI-driven system design. Shaped an HCI research identity focused on multilingual UX, CSCW, and inclusive collaboration technologies.

Louisiana State University

Teaching Assistant for CS 4444 (Artificial Intelligence)

Teaching Assistant for CS 4402 (Database Management System)

Baton Rouge, Louisiana

Jan 2024 – Aug 2024

Aug 2023 – Dec 2023

PROJECTS

PolyFlux: AI-Facilitated Multimodal Brainstorming Canvas

Baton Rouge, Louisiana

Team lead and Developer

Jan 2025

- Built an infinite-canvas collaboration tool where teams ideate with text, images, sketches, and audio while AI helps them diverge, converge, and make sense of ideas.
- Engineered a real-time whiteboard with TLDraw + React, custom shapes (including an inline audio recorder/player), and WebSocket sync for low-latency multi-user editing.
- Developed a Python Flask AI backend (LangChain) that orchestrates specialized assistant roles (e.g., Catalyst, Provocateur, Communicator, Mediator) to suggest prompts, cluster ideas, and summarize progress without overriding user agency.
- Implemented adaptive “AI nudges” that respond to phase signals (e.g., add vs. arrange rate, cluster growth, spatial entropy) to support either exploration or synthesis at the right moment.
- Project repo: <https://github.com/SaramshGautam/polyflux.git>

GitNav: AI-Powered GitHub Discussion Summarizer

Baton Rouge, Louisiana

Co-Creator & Developer

Apr 2025

- Developed a tool to generate structured summaries and timeline visualizations of GitHub Discussions and Issues using fine-tuned large language models.
- Fine-tuned LLaMA-2 and LLaMA-3 models on GitHub issue threads using LoRA and 4-bit quantization to optimize for memory efficiency and summarization quality.
- Built a full-stack web application integrating the GitHub GraphQL API with Python Flask and React to automate thread retrieval, preprocessing, and visualization.
- Implemented interactive D3.js visualizations to display discussion timelines and conversational dynamics, aiding comprehension of thread evolution.
- Project repo: <https://github.com/SaramshGautam/gitnav.git>

LINC: Language-INdependent Collaboration Tool

Baton Rouge, Louisiana

Team Lead

Aug 2024

- Developed a real-time multilingual transcription and post-meeting analysis tool to support collaboration among ESL students in academic settings.
- Integrated Whisper and WebSockets for real-time audio transcription with Python Flask backend.
- Utilized the OpenAI API for automatic translation, meeting summaries, and action items extraction from transcripts.
- Designed and implemented interactive data visualizations using D3.js for stream graphs and chord diagrams, and Chart.js for radar charts, enhancing the post-meeting analysis.

- Project repo: <https://github.com/SaramshGautam/linc.git>

Brain Tumor Detection (Python, CNN)**Kathmandu, Nepal***Team Lead**May 2021*

- Created a convolutional neural network (CNN) model to predict the likelihood of a brain tumor.
- Developed a web-based user interface using HTML, CSS, and JavaScript.
- The software uses the trained model to predict the chance of having a brain tumor.

Steganography Project (C#, Image Processing)**Kathmandu, Nepal***Team Lead**Jun 2020*

- Developed software that hides data within images to ensure the security and privacy of information.
- Utilized AES encryption for enhanced protection.
- Achieved steganography using the Least Significant Bit (LSB) algorithm.

ACTIVITIES AND LEADERSHIP

Nepalese Student Organization at LSU**Baton Rouge, Louisiana***President**Aug 2024 – Aug 2025*

- Leader of an organization of 100+ members, hosting cultural and social events, and managing a committee of executives.